

Capstone Project Topics

Traffic Forecast

- **Goal:** To predict traffic volume(or congestion levels, travel time) for specific locations or routes within a city or area
- **Data source:**
 - City's traffic data
 - Google map API
- **Model:**
 - Time series models(SARIMA)
 - LSTM
 - Spatiotemporal Models(neural network)

Energy Consumption Forecast

- **Goal:** To predict future energy consumption levels(electricity, gas, or other energy)
- **Data source:**
 - Department of Energy (DOE)
 - UCI Machine Learning Repository (for household)
- **Model:**
 - Time series models(SARIMA)
 - LSTM Neural Network

Online Ad analysis

- **Goal:** To predict future energy consumption levels(electricity, gas, or other energy)
- **Data source:**
 - Third party API: SpyFu, SEMrush
 - Google Analytics/ Google Ad API (?)
 - Meta Ad API (?)
- **Model:**
 - Regression
 - Stacking/boosting models